UVA DATA SCIENCE

Detecting Price Anomalies and Benchmarking Variability in Public Sector Procurement Data

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Presentation Roadmap

1. **Purpose and context**
2. Data and methodology
3. Notable Metrics
4. Limitations and implications
5. Recommendations and next steps

Project Purpose

Build a defensible way to compare public-sector prices before attempting to flag anomalies

The problem

- Public agencies may pay very different prices for seemingly similar goods and services
- Some variation is legitimate, while some reflects inconsistent data, unusual transactions, or prices worth reviewing
- Messy descriptions and inconsistent units make simple comparisons unreliable

Our objective

Develop and evaluate a practical analytical framework that:

- Normalizes quantities and units so prices are comparable
- Groups genuinely similar purchases using item descriptions and structured fields
- Measures price dispersion across agencies, vendors, geographies, and time
- Flags potential anomalies while clearly separating signals from data-quality artifacts
- Explains where the methodology can support real procurement decisions and where human review remains necessary

Background

Why price benchmarking in procurement data is harder than it first appears

1 Wide price dispersion

Prices can vary because of specifications, packaging, contract terms, geography, timing, vendor relationships, and purchasing volume

2 Inconsistent product language

Short descriptions often contain abbreviations, codes, omitted details, and different wording for the same item

3 Uneven data quality

Unit-of-measure fields, categories, and quantities may be missing or inconsistent, creating false comparisons and artificial outliers

Core analytical principle: a transaction should not be labeled anomalous until the team has established that it is comparable to the benchmark group.

Sponsor Background

GovSpend and the project's practical relevance

GovSpend

Government procurement intelligence platform

GovSpend works with public purchasing data to help users understand government buying activity and identify actionable procurement insights.

Project sponsor

Matt Beard

Director of Technology & Automation,
GovSpend

Sponsor-defined emphasis

- Prioritize analytical insight and methodology over production-ready tooling
- Explain what works, what does not, and why
- Document assumptions and risks of interpreting public procurement prices
- Identify potential applications for GovSpend and its public-sector customers

Data and Methodology

Final Data Scope

- Public Virginia procurement records obtained through GovSpend
- Multiple export files merged into a single line-item dataset
- Analysis limited to transactions with valid Item Category and Unit of Measure (UOM)

Preprocessing Workflow

- Merged monthly procurement exports
- Removed Records with missing UOM and Item Category
- Standardized UOM labels (e.g., EA, Each, EACH)
- Converted price fields to numeric format
- Generated summary datasets grouped by Category + UOM

Unit Normalization Approach

- Compared purchases only within the same Unit of Measure
- Evaluated package-level normalization using quantity and text parsing
- Used standardized units to improve price comparability

Item-Matching Methods

- Structured grouping using Item Category and UOM
- Evaluated UNSPSC Commodity hierarchy as an alternative grouping strategy
- Planned rule-based parsing of item descriptions to identify package sizes and equivalent products

Evaluation Sample

- Computed descriptive statistics for each Category + UOM group
  - Count
  - Mean / Median Price
  - Standard Deviation
  - Variance
  - Coefficient of Variation (CV)

Anomaly Detection Techniques

- Used the Coefficient of Variation (CV) to measure relative price dispersion across comparable purchases
- Identified high-variance categories as candidates for additional review rather than direct evidence of anomalous pricing

Notable Metrics

Table 1: Normalization

Total Rows Loaded	Resolved Rows	Declined Rows	Coefficient of Variance (Raw)	Coefficient of Variance (Normalized)
917321	43130	2865	0.653	0.078

Table 2: Hand Labeled Set Comparison

Total Rows	Numbers Provided	Marked Unknowable	High Precision	Medium Precision	Unverifiable Rows	Decline Quality	Parser and Labeler Conflicts
300	252	46	102	34	14	150	2

Limitations, Applications, and Recommendations

Honest boundaries on the method, and where it is already useful

The method decides which purchases deserve a look. A person decides what they mean.

Where a description is ambiguous, the method declines to answer rather than guesses. That choice keeps every conversion defensible, and it is why the output is a review queue rather than a verdict.

1	**Limits**	Most descriptions never state a package size. The category and unit-of-measure fields are both less reliable than expected.
2	**Uses**	Compare prices per item across agencies. Group comparable products by description. Rank what to review first.
3	**Next**	Stop grouping on category codes. Screen each-coded records. Set thresholds by review capacity, not by a target number.

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Questions